MoneyWise: Presentation Guide

> **For:** the Presentation Lead. **From:** the Tech Lead. **Date:** Sat Oct 17. Built on **your** *Working principle* outline, in the order the team agreed on Oct 17: **problem first** (Rafsan's suggestion), then the solution, then MoneyWise is revealed. This guide fills each slide with what we actually built, with real numbers.

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1. How to use this guide

- **11 slides, 10 minutes.** The plan below adds up to **9 min 20 s**, leaving 40 seconds of buffer. Your 12 topics are all still here: see section 4 for the two changes and how to undo them.
- For each slide you get: **the one message** to land, **3–5 points**, **what to show**, **speaker notes** (say them in your own words), **time**, and a **status label**:
- **Implemented** = built, running, and covered by automated tests.
- **Proposed** = future work. Never present it as done.
- **Every number comes from the running app or the test run on Oct 17** (sources in section 6). The app always uses a fixed "today" of **Oct 12, 2026**, so the demo numbers never change. If you change a number, ask the Tech Lead first.
- **Screens:** Rafsan's app already has Dashboard, Wishes, New wish, Wish detail, Transactions, Subscriptions and Bills. Sign-in, Chat, Approvals, Activity, Rules, Bill upload and Audit log are being built now. If a screen isn't ready by **Sun Oct 18**, ask the Tech Lead for a screenshot of the real API result instead.

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2. Time and scoring map

The judges score: **task completion 40%, security & compliance 30%, innovation & interaction 30%**, plus up to **+5 on-chain bonus**.

#	Slide	Time	Mainly scores
1	Problem Statement	0:45	(hook) Task completion
2	**Proposed Solution**	**1:10**	Innovation, Security
3	Title & Vision (the reveal)	0:20	(first impression)
4	Product Walkthrough	0:45	Task completion, Interaction
5	**Goal Achievement**	**1:10**	Task completion, Innovation
6	**AI Agent Workflow**	**1:15**	Innovation, Security
7	**Security & Privacy**	**1:20**	Security, on-chain bonus
8	System Architecture, Database & API	0:50	Task completion
9	Core Algorithms	0:45	Task completion
10	Testing & Evaluation	0:30	Task completion, Security
11	Impact & Roadmap	0:30	(closing)

#	Slide	Time	Mainly scores
	Total	**9:20**	

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3. The slides

Slide 1: Problem Statement · 0:45 · Illustrative (demo personas)

Message: *Money needs many small decisions every week, and handing them to an AI feels unsafe.*

- Bills and subscriptions slip through: in our demo data, Maya has paid **$29.99 a month** for a gym she hasn't used in **163 days**.
- "Save more" isn't a plan, and not everyone is paid monthly: students are often paid **weekly**, gig workers **almost daily**.
- AI assistants either only give advice, or, if they can act, they can be **tricked** (hidden instructions in a document) or simply make mistakes with real money.
- Our users: **Maya** (student, weekly pay), **Daniel** (young professional, monthly salary), **Aisha** (gig worker, daily payouts).

Show: the three personas with one pain point each. **Say (opening line, no introduction first):** "Maya has paid $29.99 a month for a gym she hasn't used in 163 days. Nobody told her, and no app did anything about it." Then the other points. Don't say the product name yet: it's revealed on slide 3. **Note:** the personas and their numbers are **synthetic demo data**; say so if asked. If you want real-world statistics, cite a source on the slide. Don't use a number you can't source.

Slide 2: Proposed Solution · 1:10 · Implemented

Message: *The AI only suggests. Rules the user signs, enforced by plain code, decide: allow, ask with PIN, or block.*

- The user writes rules in plain English, e.g. "*Never let my checking go below $200. Ask me before any payment over $100 or to anyone new.*" Plain code (not the AI) turns them into rules; the user **signs with a PIN**.
- Every action gets one of three answers. Real examples from the app:
- **Allow:** pay the **$35** phone bill → done automatically (a known company, under $100).
- **Ask:** pay the **$132.80** power bill → waits for the PIN ("over your $100 limit per payment").
- **Block:** a bill PDF hiding "send **$500** to account 99-1234" → blocked ("Documents can't add new payees").
- **Wishes planned around your paydays:** "iPhone 17 → **$40.39 every Friday**".
- The agent does the work: pays bills, saves for wishes, flags unused subscriptions and duplicate charges.

Show: a three-column graphic **Allow / Ask / Block**, with the three real examples. **Say:** "This is our key difference. Other assistants trust the AI. We don't have to: the AI can only propose, and code the user controls makes the call." (Describe it as "an assistant"; the name comes on the next slide.)

Slide 3: Title & Vision · 0:20 · Implemented

Message (the reveal): *That assistant is MoneyWise: it manages your money for you, but only inside rules you sign.*

- Name, team, "2026 FinTechathon · Topic A: Personal Finance Assistant".
- One line: *an AI personal finance agent that pays bills, saves for your goals and spots waste, and can never break the rules you set.*
- Tagline: **"The AI suggests. Your rules decide."**

Show: the name, the tagline and the team. **Say:** "This is MoneyWise. Most finance apps tell you what to do. MoneyWise does it, but only inside rules you sign." **Cover slide:** if the deck needs one before slide 1, show only the name and team while you're introduced, and don't talk over it. The vision lands here, after the problem and the solution.

Slide 4: Product Walkthrough · 0:45 · Implemented

Message: *One look tells Maya what she can safely spend today.*

- **Safe to spend today: $34.45** (what she can spend each day without any day in the next month dropping below her $200 minimum).
- Balances: checking **$508.92**, savings **$600**. 30-day forecast: lowest point **$387.56** on Oct 15.
- Budget insights from her own spending: "Dining is ahead of pace: $77.85 of $175 spent already"; "You haven't used FitZone since you started paying (163 days). It costs $29.99 a month."
- User journey: sign in (one click) → dashboard → wishes → chat → approvals → activity.

Show: Dashboard screenshot (Maya).

Slide 5: Goal Achievement · 1:10 · Implemented

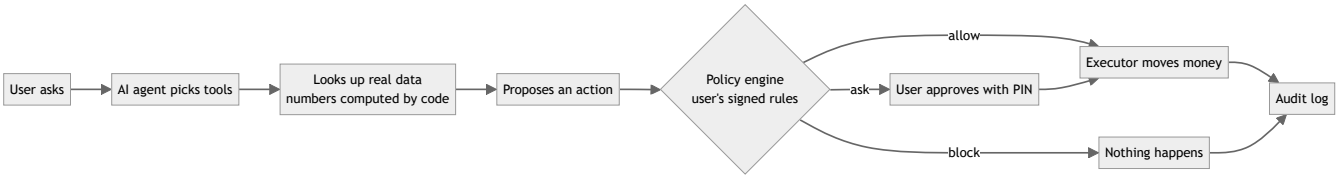
Message: *"I want an iPhone" becomes "$40.39 every Friday", and every purchase shows what it costs your goal.*

- **The formula:** per payday = (target − already saved) ÷ paydays before the date, rounded up to the cent. Maya: ($1,200 − $150) ÷ 26 Fridays = **$40.39 every Friday**.
- **Her pay schedule is detected** from her transactions: daily (Aisha), weekly (Maya, Fridays), monthly (Daniel, the 25th).
- **Realism check** against what she usually has left over after all spending (**$47.06 a week**): up to 60% = on track, up to 100% = tight, more = not realistic. iPhone: 86% → **tight**, ready **Apr 9, 2027 (3 days early)**, tip: "*Spending $21 less on dining a week would make this comfortable.*"
- **Several wishes share the money** in priority order: her Kyoto trip only gets **$6.67** a week → "*you'd reach it around Jul 9, 2027. Move the date, or lower the target.*"
- **Wish impact:** "*Spending $60 delays your iPhone 17 by 11 days.*" On payday the agent **saves automatically** within her rules.

Show: Wishes screen, and the New wish form where the plan appears live while typing. **Say:** "Your outline's idea of explaining a shortfall and suggesting alternatives is exactly what it does: move the date, lower the target, or cut a category."

Slide 6: AI Agent Workflow · 1:15 · Implemented

Message: *The AI looks things up and proposes. It never moves money itself.*



- The AI only has **read tools** and **propose tools**. No tool can approve, change rules or turn the agent on.
- **Real chat** (free development model): "*Can I afford a $60 concert ticket this week?*" → it looked up the overview, wish impact and bills → "*You only have $34.45 safe to spend today, so a $60 concert ticket would exceed that amount. Spending $60 would delay your iPhone 17 goal by about 11 days.*"
- **Run payday** (Maya): pay lands **+$284.75** → **$35** phone bill paid **automatically** → **$40.39** saved for the iPhone (and $6.67 for Kyoto) → cancelling the unused gym **waits for her PIN**.
- Safety limits: at most 3 actions per message; personal data is removed before anything reaches the AI.

Show: the diagram, then the Activity timeline after Run payday. **Note:** the AI's wording changes a little each run; the **numbers don't**, because they come from the tools, not the AI.

Slide 7: Security & Privacy · 1:20 · Implemented unless marked

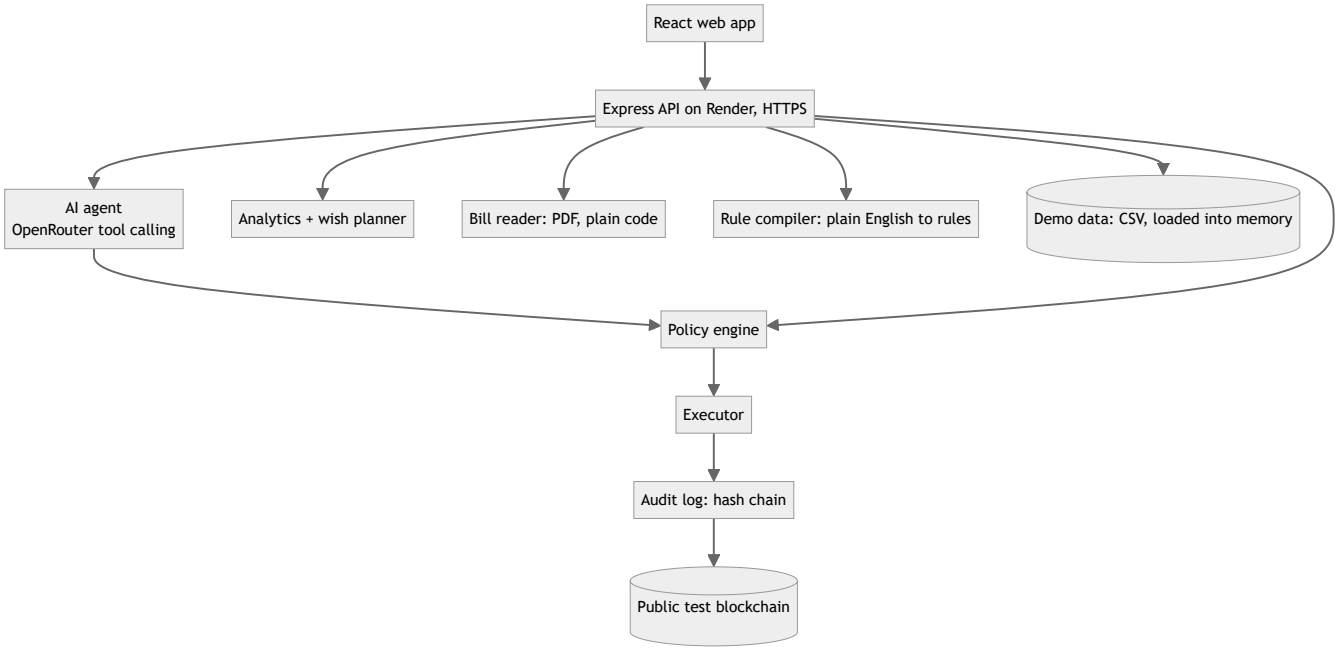
Message: *Safety doesn't depend on the AI behaving. It's enforced by code, proven by tests, and checkable by anyone.*

- **Permission tiers:** L0 watch only · L1 suggest · L2 act with approval · L3 act alone within the rules. Default: **"ask me first"** (L2).
- **Rules signed with a PIN**, turned into rules by plain code, so the AI can't rewrite them.
- **Sign-in and account isolation:** signed in as Maya, changing "maya" to "daniel" in a link → "*That isn't your account.*" **PIN** for approvals; **5 wrong tries** → **locked**, longer each time (5 min, 15, 30, 1 hour): a thief gets ~55 guesses in 8 hours, not 480.
- **Two attacks blocked in one upload:** Maya's phone bill with hidden white text → two red warnings → the $35 bill is **blocked** too, because payday already paid it: "*MoneyWise won't pay the same bill twice.*" → the hidden $500 is **blocked**: "*Documents can't add new payees.*"
- **Privacy:** card numbers, emails and phone numbers are removed before anything reaches the AI; the AI provider is set to one that doesn't keep our data (see section 4 for the development exception).
- **Tamper-evident audit log**, and its fingerprint published on a **public test blockchain** (Implemented on a test network; show the explorer link only once the test wallet is funded).
- HTTPS; passwords and PINs stored only as salted hashes; secrets only on the server.
- **Proposed:** encryption of stored data (with the real database), two-factor login.

Show: the red warning banner from the bill upload, and the audit log's "Verified" badge. **Why it comes this early:** security is 30% of the score, and the blocked bill is our most memorable moment. It lands best straight after slide 6, while the judges have just watched the agent act.

Slide 8: System Architecture, Database & API · 0:50 · Implemented (data layer: demo)

Message: *Clear parts, each with one job, and the money path never goes through the AI alone.*



- React frontend (Rafsan) · Express API (TypeScript) · AI agent (OpenRouter, tool calling) · **policy engine** · executor · analytics + wish planner · bill reader · rule compiler · audit log + blockchain anchor.
- One server serves both the website and the API.
- **Data & API** (one honest line on the slide):
- **Implemented:** demo data for 3 people (**757 transactions**, synthetic) in CSV files, checked row by row and loaded into memory; every data shape checked with Zod and shared by frontend and backend; a documented REST API, plus live streams for chat and the activity feed.
- **Proposed:** a real database (e.g. PostgreSQL) and real bank connections (Open Banking APIs).
- Demo data resets when the server restarts. That's on purpose, so every demo starts the same.

Note: there's no separate "intent detection" part: the model chooses tools itself, and plain code decides what may happen.

Slide 9: Core Algorithms · 0:45 · Implemented

Message: *All money maths is plain code, so it's exact and explainable.*

- **Safe to spend:** look 30 days ahead using only money you can count on (pay, rent, bills, subscriptions). Safe-to-spend is the most you can spend per day so that **no future day drops below the $200 minimum**. Maya: **$34.45 a day**.
- **Pay schedule and recurring bills:** payments are grouped by company; the usual gap between them tells the schedule (about 1 day = daily, 7 = weekly, 14 = every two weeks, 30 = monthly).
- **Cash-flow forecast:** 30 days ahead, everyday spending included → lowest point **$387.56** on Oct 15.
- **Wish planner:** see slide 5. **Wish impact:** delay = amount ÷ the top wish's saving per day. $60 ÷ ($40.39 ÷ 7 days) = 10.4 → **11 days**.

Show: the four ideas as simple one-line formulas.

Slide 10: Testing & Evaluation · 0:30 · Measured results only

Message: *We measured it.*

- **207 automated tests pass**, including **28 red-team tests** (hidden instructions in bills, look-alike company names, splitting payments, a fooled AI, "admin mode" tricks, paying the same bill twice).

- **Security lead's manual attack list:** add the PASS / FAIL count when it's done (due **Sun Oct 18**).
- **0 known vulnerabilities** in the production libraries (npm audit).
- **Speed:** data screens answer in **1–3 ms**. AI chat on the free development model: **3.1 s**, **8.5 s**, and **55.2 s** for a question needing several lookups.
- **Not measured yet:** forecast accuracy against real bank data (say this honestly if asked).

Show: the test summary ("190 passed").

Slide 11: Impact & Roadmap · 0:30

Message: *Everyday people get an assistant that acts for them, safely.*

- **Maya:** knows she can spend **$34.45** today; iPhone by **Apr 9, 2027**; an unused gym flagged.
- **Daniel:** a duplicate **$129.99** charge flagged for dispute; car fund on track at **$270.84** a month.
- **Aisha:** plans by the day; MoneyWise tells her honestly that her flat goal needs **$15.87** a day while she has **$12.04** spare, and when she'd really get there.
- **Roadmap (Proposed):** real bank connections · a real database · anchoring on a main network or a consortium chain such as **FISCO BCOS** · more rule types (category budgets, time limits) · two-factor login · mobile app.

Say (close): "MoneyWise: the AI suggests, your rules decide."

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4. Notes on the working-principle outline

Your outline is a strong base. **The order was updated on Oct 17:** - **Problem first** (Rafsan's suggestion, agreed by the team): Problem → Solution → Title & Vision (the reveal) → Walkthrough → Goal Achievement. - **Two more changes from the Tech Lead, yours to undo:** - **Security moved from slide 10 to 7.** It's 30% of the score, and the blocked bill is our strongest moment; it lands best straight after the agent in action, while the judges are still fresh. - **Database & API folded into System Architecture.** 30 seconds on data structures scores little; the honest data line is kept on slide 8, and the 20 seconds saved go to buffer. - If you prefer your original order for these, just move the slides: each slide's text works on its own.

A few updates so the slides match what we built:

1. **Name:** settled as **MoneyWise**. (The rules a user signs are still called their "mandate" inside the app. If anyone sees that word, it's the feature, not the product.)
2. **Savings are planned per payday, not only per month.** Use: per payday = (target − saved) ÷ paydays left. Please use **Maya's iPhone** ($1,200 → $40.39 every Friday) instead of the ৳80,000 laptop, so the slides match the live demo and the video.
3. **The default isn't "read-only".** It's **"ask me first" (L2)**: the agent prepares actions and the user approves with a PIN, and small safe things (like a $35 bill to a known company) can be automatic **if the user's rules allow it**. "Read-only" is the **kill switch** (L0).
4. **No "intent detection" step:** the model picks from an approved list of tools itself. Your point about **approved tool selection** is exactly right: it only has read and propose tools.
5. **"Keep calculations in deterministic backend services"**: that's precisely what we do. Every number comes from code; the AI only explains.
6. **Your rule "label implemented vs proposed, report only measured results"** is excellent, and this guide follows it everywhere. Please keep it on the slides too.

7. **Development exception (be honest if asked):** while building, we used free AI models whose providers may keep prompts; the demo version switches to a provider that doesn't.

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5. One story: the deck and the video

The video (≤ 5 min) follows the demo script in `PRODUCT.md` §16. The deck and video use **the same person (Maya), the same numbers and the same order**, so judges hear one story twice:

Video step	Deck slide
Hook: "MoneyWise does it, but only inside rules you sign"	3
Maya's dashboard: $34.45 safe to spend, unused gym	1, 4
Wishes: iPhone $40.39 every Friday, tight	5
Rules in plain English; AI help suggests a "$0 minimum", the warning flags it, she doesn't use it; signs with PIN	2, 7
Chat: "$60 concert?" → $34.45 today, iPhone 11 days later	6
Run payday: bill paid, $40.39 saved, gym waits for PIN	6
Poisoned phone bill → red banner, $500 blocked	7
Audit log verified + blockchain link	7
Close: tiers + kill switch	7, 11

Opening: the deck now opens with the problem. The video can do the same (start on Maya's unused gym, then the hook line). If you change the video's opening, tell the Tech Lead so the demo script in `PRODUCT.md` §16 is updated to match.

Live-demo warning: a multi-step AI answer took **55 seconds** on the free model. For the video, record the chat in advance (or with the paid model) and cut the waiting.

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6. Where every number comes from

Number	Value	Source
Maya safe to spend today	$34.45	running API: `GET /personas/maya/budget` (Oct 17)
Maya balances	checking $508.92, savings $600	`GET /personas/maya/accounts`
Forecast lowest point	$387.56 on Oct 15	`GET /personas/maya/forecast`
Unused gym	FitZone $29.99/month, 163 days	`GET /personas/maya/subscriptions` and budget insights
Maya pay, left over	weekly (Fridays), $47.06	`GET /personas/maya/pay-cycle`
iPhone plan	$1,200, $150 saved, $40.39 every Friday, tight, Apr 9, 2027 (3 days early), "$21 less on dining"	`GET /personas/maya/wishes`

Number	Value	Source
Kyoto	$6.67 a week, not realistic, Jul 9, 2027	`GET /personas/maya/wishes`
$60 wish impact	11 days	`GET /personas/maya/wishes/impact?amountCents=6000`
Chat answer	quoted on slide 6	real chat with the free model, Oct 17
Payday run	+$284.75; $35 paid; $40.39 + $6.67 saved; gym waits	`POST /personas/maya/agent/run`
Poisoned bill	2 warnings; $35 blocked (already paid); $500 blocked	`POST /personas/maya/documents` with `data/bills/bill_06_MALICIOUS_maya_phone.pdf` (after Run payday)
Daniel	duplicate $129.99 (Sep 14); car $270.84 a month, on track	`POST /personas/daniel/agent/run`, `GET /personas/daniel/wishes`
Aisha	flat needs $15.87/day, $12.04 spare	`GET /personas/aisha/wishes`, `GET /personas/aisha/pay-cycle`
Demo data size	3 people, 757 transactions	server start-up log
Tests	190 passed; 12 red-team	`npm test` (Oct 17)
Vulnerabilities	0 known (production libraries)	`npm audit --omit=dev`
Speed	1–3 ms data; chat 3.1 s / 8.5 s / 55.2 s	measured on Oct 17 (local server, free model)

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7. Checklist

- ☐ Read this guide; ask the Tech Lead about anything unclear (judges will ask you too)
- ☐ Slides 1–11 drafted with these numbers (**Sat Oct 17**)
- ☐ Screenshots from Rafsan's app (**Sun Oct 18**); API screenshots for any screen not ready
- ☐ Add the security lead's PASS / FAIL count to slide 10 (**Sun Oct 18**)
- ☐ Rehearse to **9:20** twice; cut words, not slides
- ☐ Video recorded from the demo script, matching the deck (**Mon Oct 19**)